

Motivation: Two Key Challenges in Long-form Audio Meeting QA

Challenge 1: Acoustic Missing

Speech LLMs often map audio → text tokens, losing **paralinguistic cues**: emotion, prosody, loudness, and tone.

Example: a “sudden loud voice at 19:00” is invisible to text-only representations.

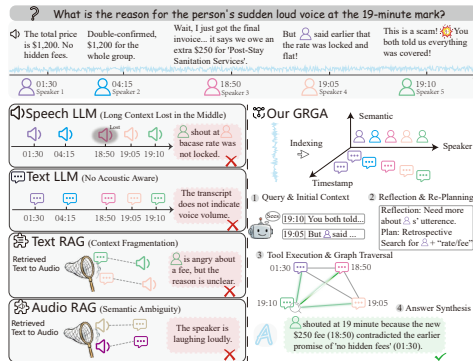
Challenge 2: Context Fragmentation

Long meetings exceed model context windows. Flat transcript RAG also struggles with **multi-hop evidence** across distant segments.

Example: linking a 1:30 description to an event at 19:00.

Our Solution

GRGA retrieves and reasons over a multi-dimensional graph;
LongAudioQA evaluates both text and acoustic grounding.



GRGA retrieves across semantic, speaker, and timestamp dimensions.